SMART BATTERY MANAGEMENT SYSTEM

Battery Condition Technical Report

REPORT ID: BMS-TEST-002
GENERATED: 2026-10-20 13:22:18 UTC

REPORTING PERIOD: 7D
BATTERY BUS: 12.0V Nominal

SPECIFICATION STATUS: UNVERIFIED PROTOTYPE · Profile: **Nominal 12V Li-ion Prototype** (Unverified). Exact chemistry and model are unverified. Provisional safety margins are applied. Do not exceed 45°C ambient or provisional voltage bounds.

SECTION A: OVERALL OBSERVED CONDITION

STATUS: WARNING

Elevated ambient temperature observed in enclosure.

SECTION B: LATEST VALID SENSOR MEASUREMENTS

Directly sampled from hardware registers via Firebase Realtime Database (/smart_bms/latest).

Parameter	Measured Value	Hardware / Path	Data Status	Technical Context / Scope
Bus Voltage	**12.45 V**	INA219 (I²C 0x40)	**VALID**	Battery terminal potential across high-side shunt.
Shunt Current	**0.920 A**	INA219 (I²C 0x40)	**VALID**	Positive: Charge / Negative: Discharge across 0.1Ω.
Active Power	**11.45 W**	Calculated (V × I)	VALID	Real-time electrical power transferred to load bus.
Ambient Temperature	**29.5 °C**	DHT11 (1-Wire D7)	**VALID**	**Enclosure ambient air** near pack (NOT cell core).
Ambient Humidity	**52.0 %**	DHT11 (1-Wire D7)	VALID	Relative air humidity inside prototype enclosure.

SECTION C: KEY FINDINGS & OBSERVATIONS

1. Voltage stable
2. Ambient temp peaked at 29.5°C

SECTION D: HISTORICAL TRENDS & STATISTICAL EXTREMA

Derived mathematically over 10 sampled interval(s) for the reporting period.

Monitored Channel	Minimum	Maximum	Average	Valid Samples	Observation / Stability
Bus Voltage	12.10 V	12.55 V	**12.32 V**	10	Bus potential within monitored bounds.
Load Current	0.50 A	1.40 A	**0.95 A**	10	Current draw across INA219 shunt.
Ambient Temperature	25.00 °C	29.50 °C	**27.25 °C**	10	DHT11 thermal buildup in enclosure.

SECTION E: EVIDENCE-BASED RECOMMENDED ACTIONS

#	Action Required	Technical Justification & Verification Target
1	**Inspect cooling fan airflow**	Ensure fan inlet is unobstructed

SECTION F: DATA QUALITY & SENSING LIMITATIONS

1. **Ambient vs. Core Thermal Measurement:** The DHT11 sensor measures enclosure air temperature adjacent to the pack. It does NOT capture internal electrochemical core cell temperature or localized thermal hotspots.
2. **Unverified Chemistry Constraints:** Exact cell chemistry and manufacturer model have not been laboratory-verified. Consequently, manufacturer-specific C-rate curves, SOC, SOH, and RUL are intentionally withheld to prevent misleading claims.
3. **Single-Node Acquisition:** Telemetry is sampled across I²C (INA219) and 1-Wire (DHT11) on the NodeMCU ESP8266 controller. Stale readings (> 60 seconds) are automatically flagged as unverified.

SECTION G: RETRIEVED TECHNICAL CITATIONS

• Texas Instruments — INA219 Bidirectional Current and Power Monitor Datasheet (SBOS448G)
• Aosong Electronics — DHT11 Humidity & Temperature Digital Sensor Technical Specification
• IEEE Std 1491-2012 — Guide for Selection and Use of Battery Monitoring Equipment in Stationary Applications
• ISO 11898-2 — Road Vehicles — Controller Area Network (CAN) — Part 2: High-Speed Medium Access Unit